DEEP LEARNING

Deep Learning

Deep Learning (DL) is a subset of **Machine Learning (ML)** that uses **Artificial Neural Networks (ANNs)** with multiple hidden layers to learn patterns from large amounts of data.

It is inspired by the structure and working of the human brain.

Deep Learning = Machine Learning + Multi-layer Neural Networks

Artificial Neural Network (ANN)

Deep learning models are built using Artificial Neural Networks**.**

Structure of ANN:

1. **Input Layer** – Receives input data
2. **Hidden Layers** – Perform computations
3. **Output Layer** – Produces final result

Each layer contains neurons (nodes)**.**

Why "Deep" Learning?

When a neural network has more than one hidden layer, it is called a Deep Neural Network **(DNN)**.

More layers → More complex pattern learning.

Example:

- 1 hidden layer → Shallow network
- 10 hidden layers → Deep network

Activation Functions

Activation functions decide whether a neuron should activate or not.

Common Activation Functions:

1. **Sigmoid**
    - Output: 0 to 1
    - Used in binary classification
2. **Tanh**
    - Output: -1 to 1
3. **ReLU (Rectified Linear Unit)**
    - Most commonly used
    - f(x) = max(0, x)
4. **Softmax**
    - Used for multi-class classification

Types of Deep Learning Networks

1. Feedforward Neural Network (FNN)

A Feedforward Neural Network (FNN) is the simplest type of Artificial Neural Network where data flows in one direction only**:**

Input Layer → Hidden Layer(s) → Output Layer

There are no loops or memory connections**.**

Key Points:

- Uses weights, bias, and activation functions
- Trained using **forward propagation** and **backpropagation**
- Mainly used for **classification and regression** problems
- Best suited for **structured/tabular data**

Example Applications:

- Student performance prediction
- House price prediction
- Fraud detection

2. Convolutional Neural Network (CNN)

A Convolutional Neural Network (CNN) is a deep learning model mainly used for image processing and computer vision tasks**.**

Key Points:

- Extracts features using **convolution layers**
- Uses **ReLU activation**
- Reduces size using **pooling layers**
- Final prediction through **fully connected layer**

Applications:

- Face recognition
- Object detection
- Medical image analysis

3. Recurrent Neural Network (RNN)

A Recurrent Neural Network (RNN) is a type of deep learning model designed for sequential data**.**

Unlike FNN, RNN has memory**,** meaning the current output depends on previous inputs.

Key Points:

- Has feedback connections
- Remembers past information
- Handles time-dependent data

Applications:

- Text generation
- Language translation
- Speech recognition
- Time series prediction

4. LSTM (Long Short-Term Memory)

LSTM (Long Short-Term Memory) is an advanced type of Recurrent Neural Network (RNN) designed to handle long-term dependencies in sequential data.

It solves the vanishing gradient problem of basic RNN.

Key Features:

- Has **memory cells**
- Uses three gates:
    - **Forget Gate**
    - **Input Gate**
    - **Output Gate**
- Can remember information for a long time

Applications:

- Language translation
- Chatbots
- Stock price prediction
- Speech recognition

5. Autoencoders

Used for:

- Dimensionality reduction
- Feature learning
- Noise removal

6. Generative Adversarial Networks (GAN)

A Generative Adversarial Network (GAN) is a deep learning model used to generate new data similar to real data.

It consists of **two neural networks**:

1.**Generator** – Creates fake data
2.**Discriminator** – Checks whether data is real or fake

Both networks compete with each other during training.

Key Points:

- Used for data generation
- Produces realistic images, audio, etc.
- Training is competitive

Applications:

- Image generation
- Deepfake
- Image enhancement
- Data augmentation

Training Process in Deep Learning

Step 1: Forward Propagation

Input → Hidden Layers → Output

Step 2: Loss Function

Measures error between predicted and actual output.

Examples:

- MSE (Mean Squared Error)
- Cross-Entropy Loss

Step 3: Backpropagation

- Error is propagated backward
- Weights are updated

Step 4: Optimization

Optimizers:

- Gradient Descent
- Stochastic Gradient Descent (SGD)
- Adam
- RMSProp

7. Hyperparameters in Deep Learning

- Learning rate
- Number of hidden layers
- Number of neurons
- Batch size
- Epochs

8. Applications of Deep Learning

1. Image Recognition
2. Speech Recognition
3. Natural Language Processing (NLP)
4. Self-driving Cars
5. Medical Diagnosis
6. Recommendation Systems

Popular Deep Learning Frameworks

- TensorFlow
- Keras
- PyTorch
- MXNet

Real-World Examples

- Google Translate
- Netflix Recommendation
- Tesla Autopilot
- Face Unlock in smartphones